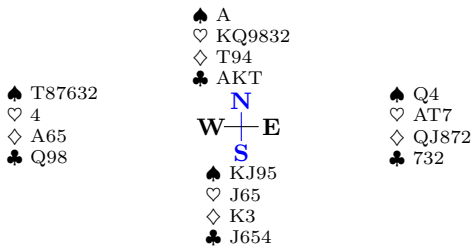
The Disclosure Gap in Contract Bridge

Research question. When AI agents learn to communicate by training together, how much of what they tell each other is missing from any written description of their system, and can training shrink that hidden part without hurting performance?

Motivation and AI safety. AI agents increasingly communicate with one another, creating a corresponding need for humans to oversee those channels. Agents trained together can develop private conventions that work for them but outsiders cannot understand. Contract Bridge, a four-player partnership card game, already regulates this problem. During bidding, partners may communicate only through legal calls, all partnership understandings must be disclosed, and unauthorized side signals are prohibited.

Current bridge AI systems achieve strong card play, yet bidding remains challenging. Systems typically follow human-designed bidding protocols or learn policies through self-play. While self-play has produced strong bidding agents, it remains unclear how to disclose their learned conventions and assess compliance with disclosure requirements.

Contract Bridge therefore provides a controlled setting for testing whether written explanations enable meaningful oversight of cooperating AI agents.



West	North	East	South
	1♡	Pass	2♡
Pass	4♡	Pass	Pass
Pass			

Figure 1: Breakdown of Bridge Board From *BridgeHand2Vec* (2023)

Methodology. I will adapt open-source bridge agents into 15–20 bidding partnerships, comparing standard self-play, varied-partner training, and a disclosure penalty, with at least three seeds per condition. Each policy will be summarized in a structured convention card.

On unseen deals, an insider trained on partnership examples and an outsider using the card will predict partner strength and suit lengths Y. Both see the same auction A, their own hand, and public context X; C denotes the card. Their predictive gap is

$$\widehat{G} = \frac{1}{n} \sum_{i=1}^{n} \log_2 \frac{q_{\text{in}}(Y_i \mid A_i, X_i)}{q_{\text{card}}(Y_i \mid A_i, X_i, C)}.$$

Positive values indicate an insider advantage in bits, not exact mutual information. A fully specified, rule-based Standard American Yellow Card (SAYC) partnership will provide a near-zero control. Omitted conventions and stronger outsiders will test validity. Evaluation will compare disclosure, playing strength, and cooperation with unfamiliar partners.

Expected contributions. I aim to establish whether insider advantages persist after improving explanations and readers, whether training reduces disclosure gaps without sacrificing score, and whether smaller gaps predict better cooperation with unfamiliar partners. The benchmark will be open-sourced to demonstrate a first step towards a clear measurement of disclosure within the Contract Bridge AI community.